เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1

Subject: 392133 Physics III

Date: 29 November 2018

Year: 2018

Section: 15-18

Time: 08.00-10.00

Name:________________________ ID:______________________ Class:__________

Instructions:

1. The examination has 7 pages (including this page), 6 problems and a total score of 60 points.
2. Write all your solution and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor' permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1.1 **[5 points]** A 160 cm woman wishes to see a full length image of herself in a plane mirror. Mr. Jeff says that the minimum length mirror required is 160 cm. Do you agree with him? Please explain.

1.2 **[5 points]** A concave spherical mirror has a radius of curvature of 24 cm. If an object is placed 18 cm in front of it, calculate the image position and draw a ray diagram to confirm your calculation.

2.1 **[5 points]** The object-lens distance for a certain converging lens is 400 mm. The image is three times the size of the object. Mr. John says that to make the image five times (the size of the object), the size of the object-lens distance must be changed to 360 mm. Do you agree with him? Please explain and show your calculation.

2.2 **[5 points]** A plano-convex glass (n = 1.5) lens has a curved side whose radius is 50 cm. If the image size is to be the same as the object size, find the object distance.

3. **[10 points]** An object is 20 cm to the left of a lens of focal length +10 cm. A second lens, of focal length +20 cm, is 60 cm to the right of the first lens. Find the distance between the original object and the final image.

4.1 **[5 points]** A far-sighted student has a near point of 75 cm. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

4.2 **[5 points]** In a Young's double-slit experiment, light of wavelength 500 nm illuminates two slits which are separated by 1 mm. Find the separation between adjacent bright fringes on a screen 5 m from the slits.

5.1 **[5 points]** Nonreflective coatings for laser lenses reduce the loss of light at various surfaces of multi-lens systems. Estimate the minimum thickness of a layer of magnesium fluoride (n = 1.38) on flint glass (n = 1.66) that will cause destructive interference of reflected light of wavelength $\lambda = 1380$ nm.

5.2 **[5 points]** A laser light ($\lambda = 3.0 \times 10^{-7}$ m) is sent through a 0.30 mm-wide single slit. What is the width of the central maximum on a screen 1.0 m from the slit?

6.1 **[5 points]** Monochromatic light is normally incident on a diffraction grating that is 2 cm wide and has 20,000 slits. The first order line is deviated at a $30°$ angle. What is the wavelength, in nm, of the incident light?

6.2 **[5 points]** Unpolarized light is passed through three successive Polaroid filters, each with its transmission axis at $45.0°$ to the preceding filter. What percentage of light gets through?

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